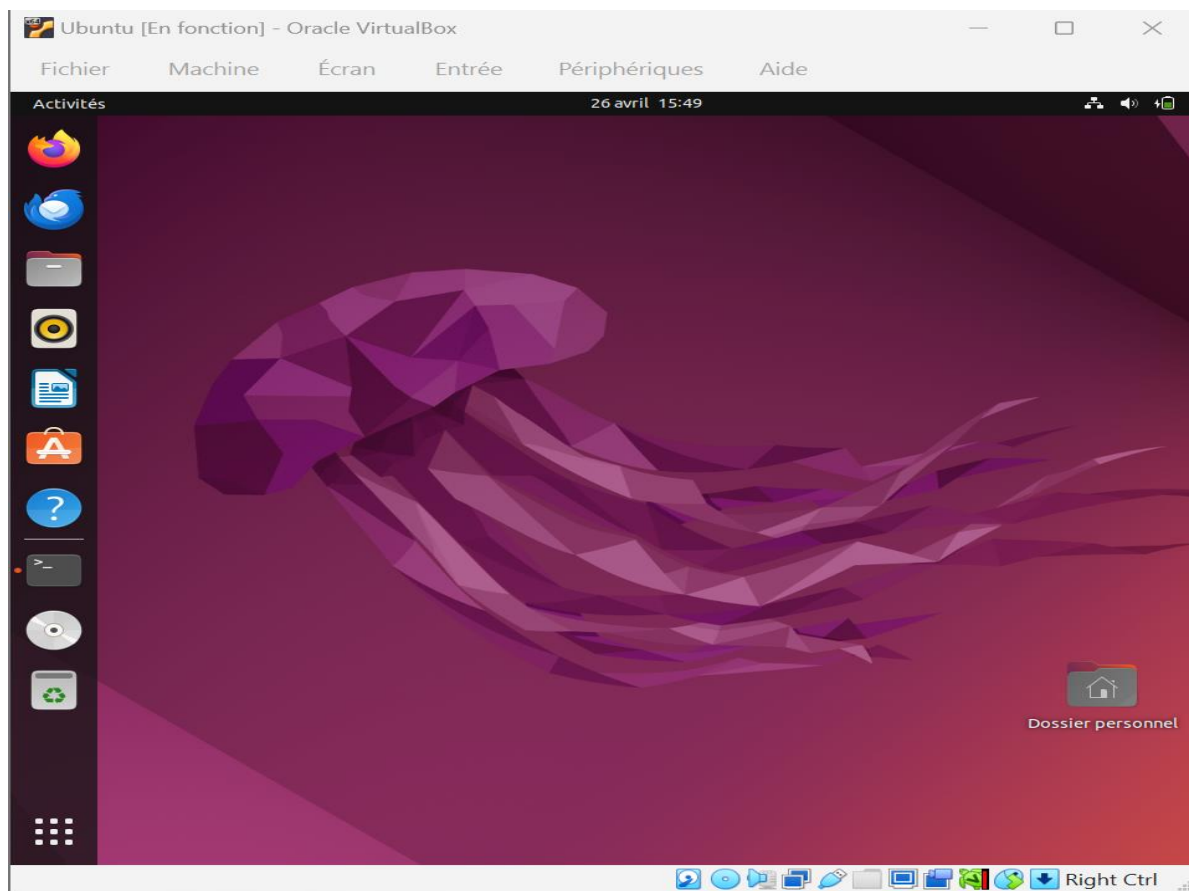


Mise en place d'une architecture LAMP distribuée

I) SERVEUR A : DNS + DHCP

Ubuntu image



1. Installer et configurer un serveur DNS

```
maimouna@maimouna-VirtualBox:~$ sudo apt install bind9 bind9utils -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
bind9 est déjà la version la plus récente (1:9.18.39-0ubuntu0.22.04.3).
Les NOUVEAUX paquets suivants seront installés :
  bind9utils
0 mis à jour, 1 nouvellement installés, 0 à enlever et 448 non mis à jour.
Il est nécessaire de prendre 3 924 o dans les archives.
Après cette opération, 62,5 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://fr.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 bind9utils all 1:9.18.39-0ubuntu0.22.04.3 [3 924 B]
3 924 o réceptionnés en 17s (234 o/s)
Sélection du paquet bind9utils précédemment désélectionné.
(Lecture de la base de données... 214867 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../bind9utils_1%3a9.18.39-0ubuntu0.22.04.3_all.deb ...
Dépaquetage de bind9utils (1:9.18.39-0ubuntu0.22.04.3) ...
Paramétrage de bind9utils (1:9.18.39-0ubuntu0.22.04.3) ...
maimouna@maimouna-VirtualBox:~$
```

```
maimouna@maimouna-VirtualBox:~$ sudo nano /etc/bind/named.conf.local
[sudo] Mot de passe de maimouna :
maimouna@maimouna-VirtualBox:~$
```

2. Créer la zone : tdsi-ucad.local

```
GNU nano 6.2 /etc/bind/named.conf.local *
zone "tdsi-ucad.local" {
    type master;
    file "/etc/bind/db.tdsi-ucad.local";
};

zone "10.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192";
};
```

Alde Écrire Chercher Couper Exécuter Emplacement Annuler Marquer -> Crochet Précédent En arrière Mot précédent
Quitter Lire fich. Remplacer Coller Justifier Aller ligne Refaire Copier Retrouver Suivant En avant Mot suivant

3. Ajouter les enregistrements pour : srv-net, srv-web et srv-db

```
maimouna@maimouna-VirtualBox:~$ sudo nano /etc/bind/db.tdsi-ucad.local
[sudo] Mot de passe de maimouna :
maimouna@maimouna-VirtualBox:~$
```

```
GNU nano 6.2 /etc/bind/db.tdsi-ucad.local *
$TTL 604800
@ IN SOA srv-net.tdsi-ucad.local. root.tdsi-ucad.local. (
    1 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS srv-net.tdsi-ucad.local.
srv-net IN A 192.168.56.10
srv-web IN A 192.168.56.20
srv-db IN A 192.168.56.30
```

4. Installer et configurer un serveur DHCP

```
maimouna@maimouna-VirtualBox:~$ sudo apt install isc-dhcp-server -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
isc-dhcp-server est déjà la version la plus récente (4.4.1-2.3ubuntu2.4).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 448 non mis à jour.
```

```
maimouna@maimouna-VirtualBox:~$ sudo nano /etc/dhcp/dhcpd.conf
[sudo] Mot de passe de maimouna :
```

5. Définir une plage d'adresses IP (ex : 192.168.56.100–150)

```
GNU nano 6.2 /etc/dhcp/dhcpd.conf
# BOOTP or DHCP.  Hosts for which no fixed address is specified can only
# be booted with DHCP, unless there is an address range on the subnet
# to which a BOOTP client is connected which has the dynamic-bootp flag
# set.
#host fantasia {
# hardware ethernet 08:00:07:26:c0:a5;
# fixed-address fantasia.example.com;
#}

# You can declare a class of clients and then do address allocation
# based on that.  The example below shows a case where all clients
# in a certain class get addresses on the 10.17.224/24 subnet, and all
# other clients get addresses on the 10.0.29/24 subnet.

#class "foo" {
# match if substring (option vendor-class-identifier, 0, 4) = "SUNW";
#}

#shared-network 224-29 {
# subnet 10.17.224.0 netmask 255.255.255.0 {
#   option routers rtr-224.example.org;
# }
# subnet 10.0.29.0 netmask 255.255.255.0 {
#   option routers rtr-29.example.org;
# }
# pool {
#   allow members of "foo";
#   range 10.17.224.10 10.17.224.250;
# }
# pool {
#   deny members of "foo";
#   range 10.0.29.10 10.0.29.230;
# }
#}

subnet 192.168.56.0 netmask 255.255.255.0 {
  range 192.168.56.100 192.168.56.150;
  option routers 192.168.56.10;
  option subnet-mask 255.255.255.0;
  option domain-name-servers 192.168.56.10;
}

^G Aide      ^O Écrire   ^M Chercher  ^X Couper   ^J Exécuter ^C Emplacement ^U Annuler  ^A Marquer  ^_] -> Crochet ^O Précédent ^B En arrière ^_ Mot précédent
^X Quitter  ^R Lire fich. ^I Remplacer ^U Coller   ^D Justifier ^_ Aller ligne ^B Refaire   ^C Copier    ^Q Retrouver ^M Suivant   ^F En avant  ^_ Mot suivant
```

```
maimouna@maimouna-VirtualBox:~$ sudo nano /etc/default/isc-dhcp-server
```

6. Configurer :

Passerelle, DNS → **serveur A**, c'est-à-dire DNS sur le serveur A ubuntu

```
maimouna@maimouna-VirtualBox:~$ sudo nano /etc/dhcp/dhcpd.conf
[sudo] Mot de passe de maimouna :
maimouna@maimouna-VirtualBox:~$
```

```
#shared-network 224-29 {
#   subnet 10.17.224.0 netmask 255.255.255.0 {
#       option routers rtr-224.example.org;
#   }
#   subnet 10.0.29.0 netmask 255.255.255.0 {
#       option routers rtr-29.example.org;
#   }
#   pool {
#       allow members of "foo";
#       range 10.17.224.10 10.17.224.250;
#   }
#   pool {
#       deny members of "foo";
#       range 10.0.29.10 10.0.29.230;
#   }
#}
subnet 192.168.56.0 netmask 255.255.255.0 {
    range 192.168.56.100 192.168.56.150;
    option routers 192.168.56.10;
    option subnet-mask 255.255.255.0;
    option domain-name-servers 192.168.56.10;
}
```

^G Aide ^O Écrire ^W Chercher ^K Couper ^T Exécuter ^C Emplacement
^X Quitter ^R Lire fich. ^_ Remplacer ^U Coller ^J Justifier ^_/ Aller ligne

Option routers : c'est l'adresse de la passerelle par défaut (souvent l'IP du NAT VirtualBox, ex. 192.168.56.10).

7. Démarrer et tester les services DNS et DHCP

a) Pour le service DNS :

```
maimouna@maimouna-VirtualBox:~$ sudo systemctl restart bind9
maimouna@maimouna-VirtualBox:~$ sudo systemctl enable named
Synchronizing state of named.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable named
maimouna@maimouna-VirtualBox:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-04-24 00:31:47 CEST; 20s ago
     Docs: man:named(8)
  Main PID: 8121 (named)
    Tasks: 4 (limit: 2269)
   Memory: 22.6M
      CPU: 136ms
   CGroup: /system.slice/named.service
           └─8121 /usr/sbin/named -u bind

avril 24 00:31:47 maimouna-VirtualBox named[8121]: zone 0.in-addr.arpa/IN: loaded serial 1
avril 24 00:31:47 maimouna-VirtualBox named[8121]: zone 255.in-addr.arpa/IN: loaded serial 1
avril 24 00:31:47 maimouna-VirtualBox named[8121]: zone localhost/IN: loaded serial 2
avril 24 00:31:47 maimouna-VirtualBox named[8121]: zone tdsi-ucad.local/IN: loaded serial 1
avril 24 00:31:47 maimouna-VirtualBox named[8121]: zone 10.168.192.in-addr.arpa/IN: loaded serial 1
avril 24 00:31:47 maimouna-VirtualBox named[8121]: all zones loaded
avril 24 00:31:47 maimouna-VirtualBox systemd[1]: Started BIND Domain Name Server.
avril 24 00:31:47 maimouna-VirtualBox named[8121]: running
avril 24 00:31:57 maimouna-VirtualBox named[8121]: managed-keys-zone: Unable to fetch DNSKEY set from 192.168.192.1:53: timeout
avril 24 00:31:57 maimouna-VirtualBox named[8121]: resolver priming query complete: timed out
lines 1-21/21 (END)...skipping...
```

Ça marche, car `bind9` est un alias vers `named`.

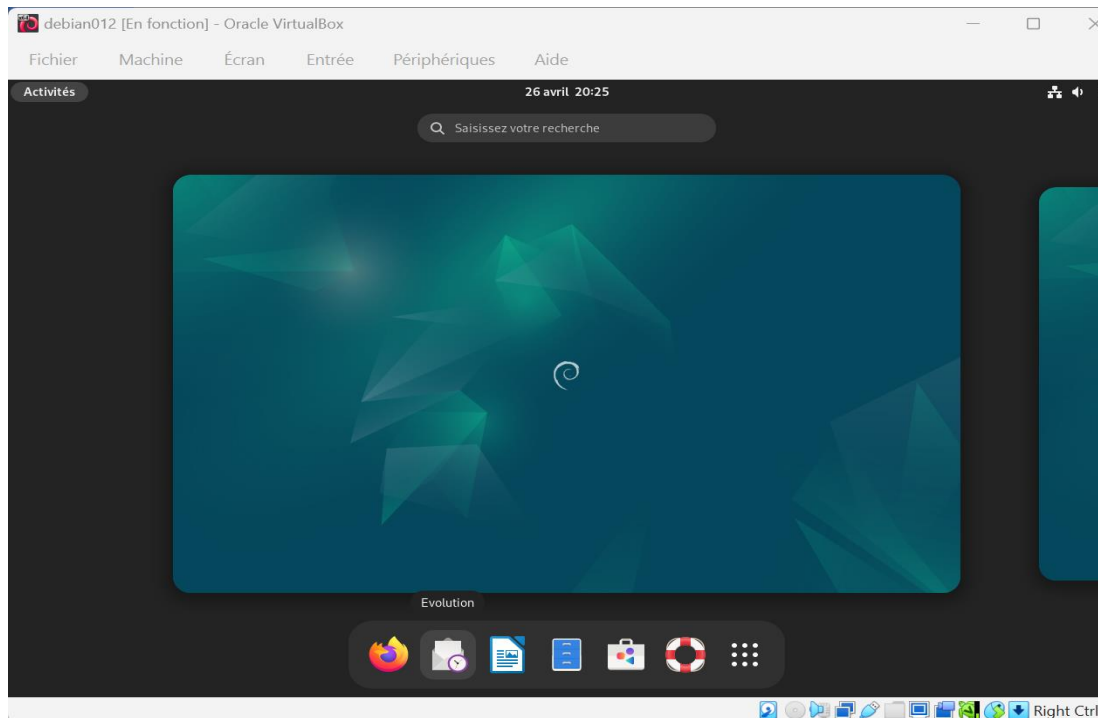
b) Pour le service DHCP

```
mainouna@mainouna-VirtualBox:~$ sudo systemctl restart isc-dhcp-server
mainouna@mainouna-VirtualBox:~$ sudo systemctl enable isc-dhcp-server
Synchronizing state of isc-dhcp-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable isc-dhcp-server
mainouna@mainouna-VirtualBox:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/lib/systemd/system/isc-dhcp-server.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-04-22 22:50:49 CEST; 8s ago
     Docs: man:dhcpd(8)
    Main PID: 5089 (dhcpd)
      Tasks: 4 (limit: 2269)
     Memory: 4.5M
        CPU: 41ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─5089 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf enp0s8

avril 22 22:50:49 mainouna-VirtualBox dhcpd[5089]: Database file: /var/lib/dhcp/dhcpd.leases
avril 22 22:50:49 mainouna-VirtualBox dhcpd[5089]: PID file: /run/dhcp-server/dhcpd.pid
avril 22 22:50:49 mainouna-VirtualBox dhcpd[5089]: Wrote 0 leases to leases file.
avril 22 22:50:49 mainouna-VirtualBox dhcpd[5089]: Listening on LPF/enp0s8/08:00:27:b3:d4:bd/192.168.56.0/24
avril 22 22:50:49 mainouna-VirtualBox sh[5089]: Listening on LPF/enp0s8/08:00:27:b3:d4:bd/192.168.56.0/24
avril 22 22:50:49 mainouna-VirtualBox dhcpd[5089]: Sending on LPF/enp0s8/08:00:27:b3:d4:bd/192.168.56.0/24
avril 22 22:50:49 mainouna-VirtualBox sh[5089]: Sending on LPF/enp0s8/08:00:27:b3:d4:bd/192.168.56.0/24
avril 22 22:50:49 mainouna-VirtualBox dhcpd[5089]: Sending on Socket/fallback/fallback-net
avril 22 22:50:49 mainouna-VirtualBox sh[5089]: Sending on Socket/fallback/fallback-net
avril 22 22:50:49 mainouna-VirtualBox dhcpd[5089]: Server starting service.
mainouna@mainouna-VirtualBox:~$ sudo nano /etc/default/isc-dhcp-server
mainouna@mainouna-VirtualBox:~$
```

II) SERVEUR B : Serveur Web (Apache + PHP)

Debian image



1. Installer Apache

```
maimouna@debian:~$ sudo apt install apache2 -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
apache2 est déjà la version la plus récente (2.4.66-1-deb12u1).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
maimouna@debian:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-04-25 03:22:59 CEST; 13min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 11046 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 11050 (apache2)
    Tasks: 6 (limit: 2277)
   Memory: 11.5M
      CPU: 283ms
   CGroup: /system.slice/apache2.service
           └─11050 /usr/sbin/apache2 -k start
             └─11052 /usr/sbin/apache2 -k start
               └─11053 /usr/sbin/apache2 -k start
                 └─11054 /usr/sbin/apache2 -k start
                   └─11055 /usr/sbin/apache2 -k start
                     └─11056 /usr/sbin/apache2 -k start

avril 25 03:22:59 debian systemd[1]: Starting apache2.service - The Apache HTTP Server...
avril 25 03:22:59 debian apachectl[11049]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 1
avril 25 03:22:59 debian systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-20/20 (END)
```

2. Installer PHP et les modules nécessaires

```
maimouna@debian:~$ sudo apt install php libapache2-mod-php -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
php est déjà la version la plus récente (2:8.2+93).
libapache2-mod-php est déjà la version la plus récente (2:8.2+93).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
maimouna@debian:~$ sudo apt install php-mysql -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
php-mysql est déjà la version la plus récente (2:8.2+93).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
maimouna@debian:~$ php -v
PHP 8.2.30 (cli) (built: Feb 28 2026 07:07:34) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.2.30, Copyright (c) Zend Technologies
    with Zend OPcache v8.2.30, Copyright (c), by Zend Technologies
```

3. Configurer le serveur pour utiliser le DNS interne

Cela force Debian à utiliser mon **Serveur A (Ubuntu : DNS/DHCP)** pour la résolution des noms.

```
maimouna@debian:~$ sudo nano /etc/resolv.conf
```

```
GNU nano 7.2 /etc/resolv.conf *
# Generated by NetworkManager
nameserver 192.168.56.10
nameserver 8.8.8.8

```

^G Aide	^O Écrire	^W Chercher	^K Couper	^T Exécuter	^C Emplacement	M-U Annuler	M
^X Quitter	^R Lire fich.	^V Remplacer	^U Coller	^J Justifier	^_ Aller ligne	M-E Refaire	M

4. Déployer une page PHP de test

```
maimouna@debian:~$ sudo nano /var/www/html/index.php
maimouna@debian:~$
```



```
maimouna@debian: ~
/var/www/html/index.php *
GNU nano 7.2
<?php
$servername = "srv-db.tdsi-ucad.local";
$username   = "admin";
$password   = "mouna1407";
$dbname     = "tdsi";
$port      = 3306;
?>
```

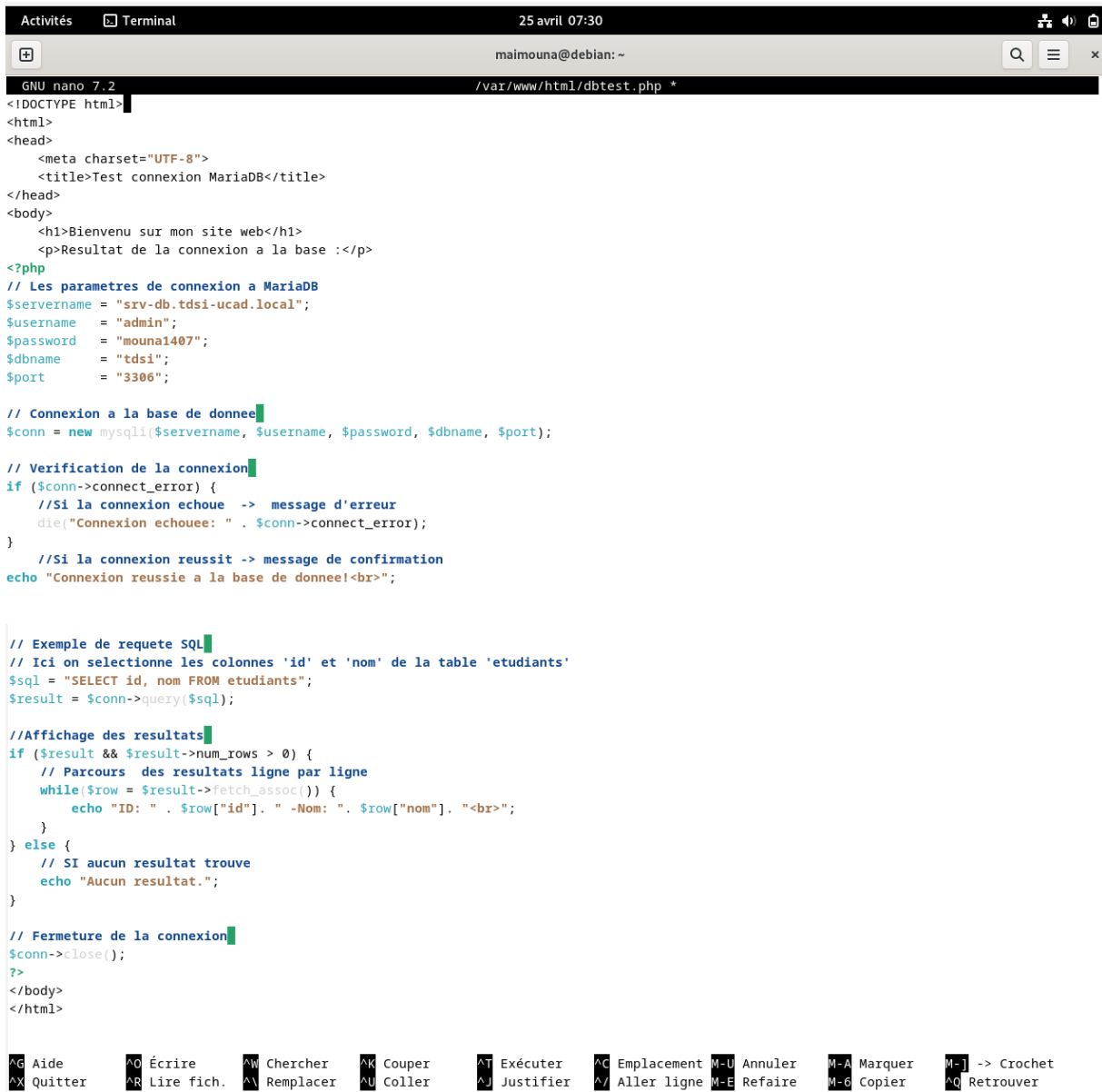
^G Aide ^O Écrire ^W Chercher ^K Couper ^T Exécuter ^C Emplacement M-U Annuler M
^X Quitter ^R Lire fich. ^\ Remplacer ^U Coller ^J Justifier ^/ Aller ligne M-E Refaire M

Rechargeons Apache

```
maimouna@debian:~$ sudo systemctl restart apache2
[sudo] Mot de passe de maimouna :
```

5. Héberger l'application simple permettant :
- de se connecter à la base de données distante (serveur C)
 - et d'afficher les informations de la base de données sur la page si la connexion est réussie.

```
maimouna@debian:~$ sudo nano /var/www/html/dbtest.php
maimouna@debian:~$
```



```
GNU nano 7.2 /var/www/html/dbtest.php *
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Test connexion MariaDB</title>
</head>
<body>
  <h1>Bienvenu sur mon site web</h1>
  <p>Resultat de la connexion a la base :</p>
<?php
// Les parametres de connexion a MariaDB
$servername = "srv-db.tdsi-ucad.local";
$username = "admin";
$password = "mouna1407";
$dbname = "tdsi";
$port = "3306";

// Connexion a la base de donnee
$conn = new mysqli($servername, $username, $password, $dbname, $port);

// Verification de la connexion
if ($conn->connect_error) {
    //Si la connexion echoue -> message d'erreur
    die("Connexion echouee: " . $conn->connect_error);
}
//Si la connexion reussit -> message de confirmation
echo "Connexion reussie a la base de donnee!<br>";

// Exemple de requete SQL
// Ici on selectionne les colonnes 'id' et 'nom' de la table 'etudiants'
$sql = "SELECT id, nom FROM etudiants";
$result = $conn->query($sql);

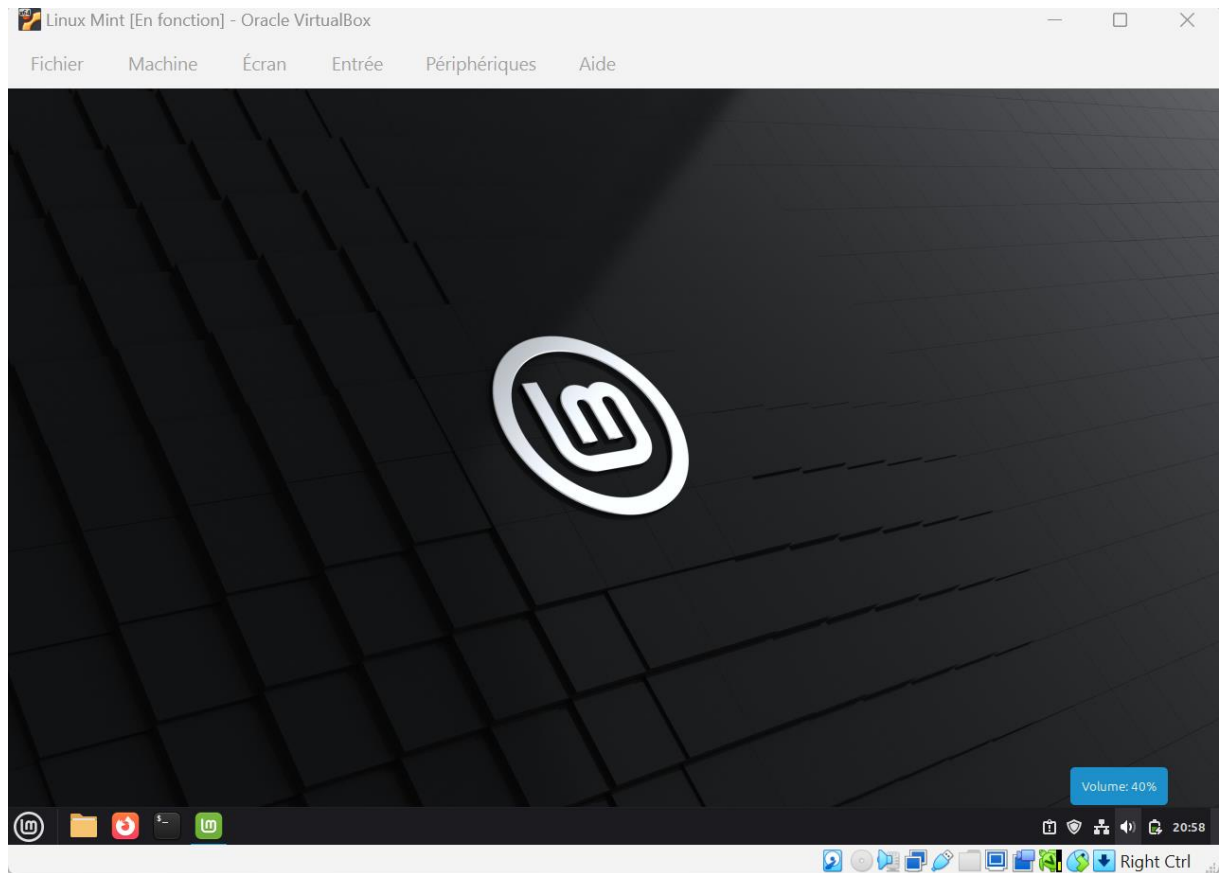
//Affichage des resultats
if ($result && $result->num_rows > 0) {
    // Parcours des resultats ligne par ligne
    while($row = $result->fetch_assoc()) {
        echo "ID: " . $row["id"]. " -Nom: " . $row["nom"]. "<br>";
    }
} else {
    // SI aucun resultat trouve
    echo "Aucun resultat.";
}

// Fermeture de la connexion
$conn->close();
?>
</body>
</html>
```

⌘ Aide ⌘ Écrire ⌘ Chercher ⌘ Couper ⌘ Exécuter ⌘ Emplacement ⌘ Annuler ⌘ Marquer ⌘ -> Crochet
⌘ Quitter ⌘ Lire fich. ⌘ Remplacer ⌘ Coller ⌘ Justifier ⌘ Aller ligne ⌘ Refaire ⌘ Copier ⌘ Retrouver

III) SERVEUR C : Base de données (MySQL / MariaDB)

Linux-Mint image



1. Installer MariaDB ou MySQL

```
maimouna@maimouna-VirtualBox:~$ sudo apt install mariadb-server mariadb-client -y
[sudo] Mot de passe de maimouna :
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
mariadb-server est déjà la version la plus récente (1:10.11.14-0ubuntu0.24.04.1).
mariadb-client est déjà la version la plus récente (1:10.11.14-0ubuntu0.24.04.1).
mariadb-client passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
maimouna@maimouna-VirtualBox:~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-04-25 23:33:56 CEST; 25min ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 1001 (mariabdb)
   Status: "Taking your SQL requests now..."
     Tasks: 9 (limit: 14865)
    Memory: 125.4M (peak: 129.9M)
       CPU: 3.460s
    CGroup: /system.slice/mariadb.service
            └─1001 /usr/sbin/mariabdb

avril 25 23:33:55 maimouna-VirtualBox mariabdb[1001]: 2026-04-25 23:33:55 0 [Note] Plugin 'FEEDBACK' is disabled.
avril 25 23:33:55 maimouna-VirtualBox mariabdb[1001]: 2026-04-25 23:33:55 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buffer_pool
avril 25 23:33:55 maimouna-VirtualBox mariabdb[1001]: 2026-04-25 23:33:55 0 [Warning] You need to use --log-bin to make --expire-logs-days or --binlog-expire
avril 25 23:33:55 maimouna-VirtualBox mariabdb[1001]: 2026-04-25 23:33:55 0 [Note] InnoDB: Buffer pool(s) load completed at 260425 23:33:55
avril 25 23:33:55 maimouna-VirtualBox mariabdb[1001]: 2026-04-25 23:33:55 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
avril 25 23:33:56 maimouna-VirtualBox mariabdb[1001]: 2026-04-25 23:33:56 0 [Note] /usr/sbin/mariabdb: ready for connections.
avril 25 23:33:56 maimouna-VirtualBox mariabdb[1001]: Version: '10.11.14-MariaDB-0ubuntu0.24.04.1' socket: '/run/mysqld/mysqld.sock' port: 3306 Ubuntu 24.
avril 25 23:33:56 maimouna-VirtualBox systemd[1]: Started mariadb.service - MariaDB 10.11.14 database server.
avril 25 23:33:57 maimouna-VirtualBox /etc/mysql/debian-start[1044]: Checking for insecure root accounts.
avril 25 23:33:57 maimouna-VirtualBox /etc/mysql/debian-start[1051]: Triggering mysam-recover for all MyISAM tables and aria-recover for all Aria tables
lines 1-23/23 (END)
```

2. Sécuriser l'installation

```
maimouna@maimouna-VirtualBox:~$ sudo mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] Y
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] Y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!
```

By default, a MariaDB installation has an anonymous user, allowing anyone to log into MariaDB without having to have a user account created for them. This is intended only for testing, and to make the installation go a bit smoother. You should remove them before moving into a production environment.

Remove anonymous users? [Y/n] Y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] Y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

Remove test database and access to it? [Y/n] Y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

Reload privilege tables now? [Y/n] Y
... Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB installation should now be secure.

Thanks for using MariaDB!

3. Créer : une base de données, schéma de BD, Insérer les données et un utilisateur avec mot de passe, avec les droits

Quand on installe MariaDB, il crée par défaut un compte **administrateur** appelé `root`

```
maimouna@maimouna-VirtualBox:~$ sudo mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 41
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE tdsi;
Query OK, 1 row affected (0,001 sec)

MariaDB [(none)]> USE tdsi;
Database changed
MariaDB [tdsi]> CREATE TABLE etudiants (
  -> id INT AUTO INCREMENT PRIMARY KEY,
  -> nom VARCHAR(100) NOT NULL
  -> );
Query OK, 0 rows affected (0,135 sec)

MariaDB [tdsi]> INSERT INTO etudiants (nom) VALUES ('Maimouna'), ('Aminata'), ('Amdou'), ('HabyLy');
Query OK, 4 rows affected (0,042 sec)
Records: 4  Duplicates: 0  Warnings: 0

MariaDB [tdsi]> SELECT * FROM etudiants;
+-----+
| id | nom |
+-----+
| 1 | Maimouna |
| 2 | Aminata |
| 3 | Amdou |
| 4 | HabyLy |
+-----+
4 rows in set (0,006 sec)

MariaDB [tdsi]> CREATE USER 'admin'@'%' IDENTIFIED BY 'mouna1407';
Query OK, 0 rows affected (0,018 sec)

MariaDB [tdsi]> GRANT ALL PRIVILEGES ON tdsi.* TO 'admin'@'%';
Query OK, 0 rows affected (0,017 sec)

MariaDB [tdsi]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0,004 sec)

MariaDB [tdsi]> exit;
Bye
```

4. Autoriser les connexions à distance depuis le serveur B

Éditons la configuration MariaDB :

```
maimouna@maimouna-VirtualBox:~$ sudo nano /etc/mysql/mariadb.conf.d/50-server.conf

maimouna@maimouna-VirtualBox: ~
GNU nano 7.2 /etc/mysql/mariadb.conf.d/50-server.conf *
bind-address = 0.0.0.0
```

Puis redémarrons MariaDB :

```
maimouna@maimouna-VirtualBox:~$ sudo systemctl restart mariadb
maimouna@maimouna-VirtualBox:~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-04-26 00:11:14 CEST; 34s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 2645 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysql (code=exited, status=0/SUCCESS)
   Process: 2648 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
   Process: 2650 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera recovery ] && VAR= || VAR=/usr/bin/galera recovery; [ $? -eq 0 ] && systemctl set-envi
   Process: 2722 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
   Process: 2724 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
  Main PID: 2710 (mariadb)
    Status: "Taking your SQL requests now..."
     Tasks: 12 (limit: 14865)
    Memory: 79.8M (peak: 82.7M)
       CPU: 2.481s
    CGroup: /system.slice/mariadb.service
            └─2710 /usr/sbin/mariadb

avril 26 00:11:14 maimouna-VirtualBox mariadb[2710]: 2026-04-26 0:11:14 0 [Note] Plugin 'FEEDBACK' is disabled.
avril 26 00:11:14 maimouna-VirtualBox mariadb[2710]: 2026-04-26 0:11:14 0 [Warning] You need to use --log-bin to make --expire-logs-days or --binlog-expire
avril 26 00:11:14 maimouna-VirtualBox mariadb[2710]: 2026-04-26 0:11:14 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
avril 26 00:11:14 maimouna-VirtualBox mariadb[2710]: 2026-04-26 0:11:14 0 [Note] InnoDB: Buffer pool(s) load completed at 260426 0:11:14
avril 26 00:11:14 maimouna-VirtualBox mariadb[2710]: 2026-04-26 0:11:14 0 [Note] /usr/sbin/mariadb: ready for connections.
avril 26 00:11:14 maimouna-VirtualBox mariadb[2710]: Version: '10.11.14-MariaDB-0ubuntu0.24.04.1' socket: '/run/mysql/mysql.sock' port: 3306 Ubuntu 24.
avril 26 00:11:14 maimouna-VirtualBox systemd[1]: Started mariadb.service - MariaDB 10.11.14 database server.
avril 26 00:11:15 maimouna-VirtualBox /etc/mysql/debian-start[2727]: Upgrading MariaDB tables if necessary.
avril 26 00:11:15 maimouna-VirtualBox /etc/mysql/debian-start[2738]: Checking for insecure root accounts.
avril 26 00:11:15 maimouna-VirtualBox /etc/mysql/debian-start[2742]: Triggering myisam-recover for all MyISAM tables and aria-recover for all Aria tables
lines 1-28/28 (END)
```

5. Adapter la configuration réseau pour permettre l'accès distant

La partie “Adapter la configuration réseau pour permettre l'accès distant” veut dire qu'il ne suffit pas d'installer MariaDB et de créer ta base mais il faut aussi que le **Serveur C (Linux Mint – MariaDB)** soit accessible depuis le **Serveur B (Debian – Apache/PHP)**.

C'est-à-dire : ajuster les paramètres réseau pour que les serveurs puissent communiquer entre eux correctement.

Vérifions que MariaDB est actif

```
maimouna@maimouna-VirtualBox:~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-04-26 01:11:02 CEST; 12min ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 4015 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCESS)
   Process: 4018 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
   Process: 4020 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= || VAR="/usr/bin/galera_recovery"; [ $? -eq 0 ] && systemctl set-envi
   Process: 4093 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
   Process: 4095 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
  Main PID: 4080 (mariadb)
    Status: "Taking your SQL requests now..."
     Tasks: 9 (limit: 14865)
    Memory: 78.5M (peak: 81.6M)
       CPU: 1.178s
    CGroup: /system.slice/mariadb.service
            └─4080 /usr/sbin/mariadb

avril 26 01:11:02 maimouna-VirtualBox mariadb[4080]: 2026-04-26 1:11:02 0 [Note] Plugin 'FEEDBACK' is disabled.
avril 26 01:11:02 maimouna-VirtualBox mariadb[4080]: 2026-04-26 1:11:02 0 [Warning] You need to use --log-bin to make --expire-logs-days or --binlog-expire
avril 26 01:11:02 maimouna-VirtualBox mariadb[4080]: 2026-04-26 1:11:02 0 [Note] Server socket created on IP: '0.0.0.0', port: '3306'.
avril 26 01:11:02 maimouna-VirtualBox mariadb[4080]: 2026-04-26 1:11:02 0 [Note] InnoDB: Buffer pool(s) load completed at 260426 1:11:02
avril 26 01:11:02 maimouna-VirtualBox mariadb[4080]: 2026-04-26 1:11:02 0 [Note] /usr/sbin/mariadb: ready for connections.
avril 26 01:11:02 maimouna-VirtualBox mariadb[4080]: Version: '10.11.14-MariaDB-0ubuntu0.24.04.1' socket: '/run/mysqld/mysqld.sock' port: 3306 Ubuntu 24.
avril 26 01:11:02 maimouna-VirtualBox systemd[1]: Started mariadb.service - MariaDB 10.11.14 database server.
avril 26 01:11:02 maimouna-VirtualBox /etc/mysql/debian-start[4098]: Upgrading MariaDB tables if necessary.
avril 26 01:11:03 maimouna-VirtualBox /etc/mysql/debian-start[4109]: Checking for insecure root accounts.
avril 26 01:11:03 maimouna-VirtualBox /etc/mysql/debian-start[4113]: Triggering myisam-recover for all MyISAM tables and aria-recover for all Aria tables
lines 1-28/28 (END)
```

Confirmons que le serveur C est joignable depuis les autres machines : Tester la connectivité réseau

```
maimouna@maimouna-VirtualBox:~$ ping 192.168.56.30
PING 192.168.56.30 (192.168.56.30) 56(84) bytes of data.
64 bytes from 192.168.56.30: icmp seq=1 ttl=64 time=0.088 ms
64 bytes from 192.168.56.30: icmp seq=2 ttl=64 time=0.042 ms
64 bytes from 192.168.56.30: icmp seq=3 ttl=64 time=0.041 ms
64 bytes from 192.168.56.30: icmp seq=4 ttl=64 time=0.048 ms
64 bytes from 192.168.56.30: icmp seq=5 ttl=64 time=0.033 ms
64 bytes from 192.168.56.30: icmp seq=6 ttl=64 time=0.033 ms
64 bytes from 192.168.56.30: icmp seq=7 ttl=64 time=0.030 ms
64 bytes from 192.168.56.30: icmp seq=8 ttl=64 time=0.031 ms
64 bytes from 192.168.56.30: icmp seq=9 ttl=64 time=0.035 ms
64 bytes from 192.168.56.30: icmp seq=10 ttl=64 time=0.031 ms
64 bytes from 192.168.56.30: icmp seq=11 ttl=64 time=0.029 ms
64 bytes from 192.168.56.30: icmp seq=12 ttl=64 time=0.041 ms
64 bytes from 192.168.56.30: icmp seq=13 ttl=64 time=0.049 ms
64 bytes from 192.168.56.30: icmp seq=14 ttl=64 time=0.037 ms
64 bytes from 192.168.56.30: icmp seq=15 ttl=64 time=0.034 ms
64 bytes from 192.168.56.30: icmp seq=16 ttl=64 time=0.035 ms
64 bytes from 192.168.56.30: icmp seq=17 ttl=64 time=0.063 ms
64 bytes from 192.168.56.30: icmp seq=18 ttl=64 time=0.036 ms
64 bytes from 192.168.56.30: icmp seq=19 ttl=64 time=0.030 ms
64 bytes from 192.168.56.30: icmp seq=20 ttl=64 time=0.041 ms
64 bytes from 192.168.56.30: icmp seq=21 ttl=64 time=0.036 ms
64 bytes from 192.168.56.30: icmp seq=22 ttl=64 time=0.036 ms
64 bytes from 192.168.56.30: icmp seq=23 ttl=64 time=0.043 ms
64 bytes from 192.168.56.30: icmp seq=24 ttl=64 time=0.044 ms
64 bytes from 192.168.56.30: icmp seq=25 ttl=64 time=0.037 ms
64 bytes from 192.168.56.30: icmp seq=26 ttl=64 time=0.030 ms
64 bytes from 192.168.56.30: icmp seq=27 ttl=64 time=0.046 ms
```

MariaDB doit écouter sur toutes les interfaces pour accepter les connexions distantes.

```
maimouna@maimouna-VirtualBox:~$ sudo ss -tlnp | grep 3306
LISTEN 0      80          0.0.0.0:*        0.0.0.0:*        users:((("mariadb",pid=4080,fd=23))
```

Testons la connexion locale

Vérifions qu'on peut se connecter à la base depuis le serveur C.

```
mainouna@mainouna-VirtualBox:~$ mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

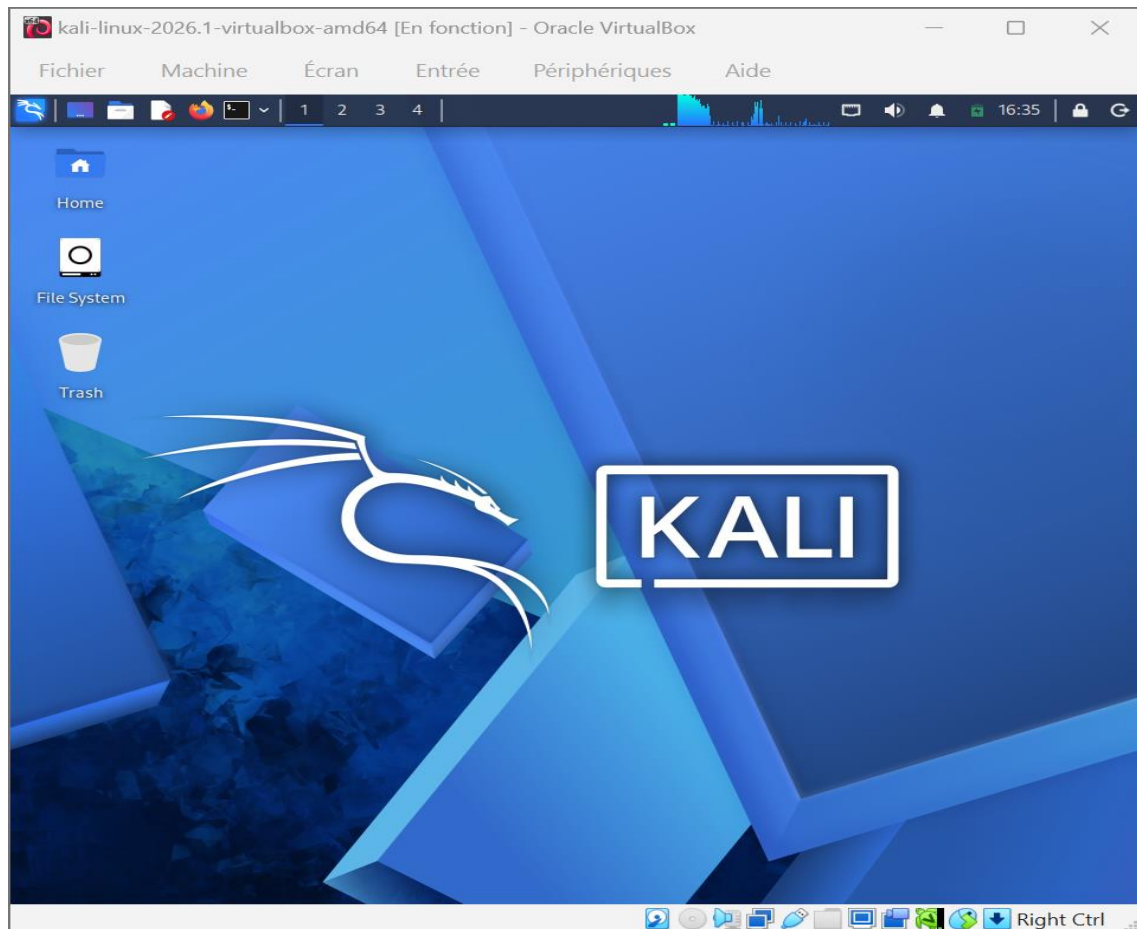
MariaDB [(none)]> USE tdsi;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [tdsi]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| tdsi |
+-----+
5 rows in set (0.001 sec)

MariaDB [tdsi]>
```

IV) MACHINE CLIENT

Kali-Linux image



1) Configurer la machine en DHCP

Configurer la machine en **DHCP** veut dire : **laisser le serveur DHCP attribuer automatiquement l'adresse IP, la passerelle et le DNS au client**, au lieu de les fixer manuellement.

```
$ sudo dhclient -v eth0
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/eth0/08:00:27:8a:35:d2
Sending on   LPF/eth0/08:00:27:8a:35:d2
Sending on   Socket/fallback
DHCPREQUEST for 192.168.56.104 on eth0 to 255.255.255.255 port 67
DHCPACK of 192.168.56.104 from 192.168.56.10
Error: ipv4: Address already assigned.
bound to 192.168.56.104 -- renewal in 239 seconds.
```

2) Vérifier l'attribution automatique d'une IP

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic eth0
        valid_lft 458sec preferred_lft 458sec
    inet 192.168.56.103/24 brd 192.168.56.255 scope global secondary dynamic noprefixroute eth0
        valid_lft 483sec preferred_lft 483sec
    inet6 fe80::cab5:fe6:5039:c085/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

3) Vérifier la résolution DNS

```
(kali@kali)-[~]
$ ping srv-net.tdsi-ucad.local
PING srv-net.tdsi-ucad.local (192.168.56.10) 56(84) bytes of data:
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=1.12 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=1.31 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=2.01 ms
64 bytes from 192.168.56.10: icmp_seq=4 ttl=64 time=1.65 ms
64 bytes from 192.168.56.10: icmp_seq=5 ttl=64 time=2.33 ms
64 bytes from 192.168.56.10: icmp_seq=6 ttl=64 time=2.32 ms
```

```
(kali@kali)-[~]
$ ping srv-web.tdsi-ucad.local
PING srv-web.tdsi-ucad.local (192.168.56.20) 56(84) bytes of data:
64 bytes from 192.168.56.20: icmp_seq=1 ttl=64 time=1.39 ms
64 bytes from 192.168.56.20: icmp_seq=2 ttl=64 time=0.768 ms
64 bytes from 192.168.56.20: icmp_seq=3 ttl=64 time=1.33 ms
64 bytes from 192.168.56.20: icmp_seq=4 ttl=64 time=1.69 ms
64 bytes from 192.168.56.20: icmp_seq=5 ttl=64 time=1.95 ms
64 bytes from 192.168.56.20: icmp_seq=6 ttl=64 time=1.39 ms
64 bytes from 192.168.56.20: icmp_seq=7 ttl=64 time=1.25 ms
64 bytes from 192.168.56.20: icmp_seq=8 ttl=64 time=1.41 ms
64 bytes from 192.168.56.20: icmp_seq=9 ttl=64 time=0.996 ms
64 bytes from 192.168.56.20: icmp_seq=10 ttl=64 time=1.86 ms
64 bytes from 192.168.56.20: icmp_seq=11 ttl=64 time=2.05 ms
```

```
(kali@kali)-[~]
$ ping srv-db.tdsi-ucad.local
PING srv-db.tdsi-ucad.local (192.168.56.30) 56(84) bytes of data.
^C
— srv-db.tdsi-ucad.local ping statistics —
712 packets transmitted, 0 received, 100% packet loss, time 756596ms
```

4) Tester la connectivité

a) Ping vers le serveur web :

```
(kali@kali)-[~]
$ ping srv-web.tdsi-ucad.local
PING srv-web.tdsi-ucad.local (192.168.56.20) 56(84) bytes of data.
64 bytes from 192.168.56.20: icmp_seq=1 ttl=64 time=1.39 ms
64 bytes from 192.168.56.20: icmp_seq=2 ttl=64 time=0.768 ms
64 bytes from 192.168.56.20: icmp_seq=3 ttl=64 time=1.33 ms
64 bytes from 192.168.56.20: icmp_seq=4 ttl=64 time=1.69 ms
64 bytes from 192.168.56.20: icmp_seq=5 ttl=64 time=1.95 ms
64 bytes from 192.168.56.20: icmp_seq=6 ttl=64 time=1.39 ms
64 bytes from 192.168.56.20: icmp_seq=7 ttl=64 time=1.25 ms
64 bytes from 192.168.56.20: icmp_seq=8 ttl=64 time=1.41 ms
64 bytes from 192.168.56.20: icmp_seq=9 ttl=64 time=0.996 ms
64 bytes from 192.168.56.20: icmp_seq=10 ttl=64 time=1.86 ms
64 bytes from 192.168.56.20: icmp_seq=11 ttl=64 time=2.05 ms
```

b) Accès HTTP au serveur web :

Accès HTTP au serveur web veut simplement dire : vérifier que ta machine cliente (Kali) peut **ouvrir la page web servie par Apache** sur ton serveur Debian (`srv-web`).

HTTP est le protocole utilisé par les navigateurs (comme Firefox, Chrome) pour afficher des pages web.

